

Prompt Engineering Workshop Outline (DRAFT)

Rutgers IDEA x JPMorganChase

1. How AI Works

Before writing better prompts, it helps to know what is actually happening when you hit Send.

The Key Technologies

Term	What it means
NLP	Helps computers understand human language. Example: London Stock Exchange (LSEG) scans earnings call transcripts to flag sentiment.
LLMs	Trained on massive text data to generate, summarize, classify, and reason across language tasks.
GenAI	Creates new content (text, slides, reports, code) using learned patterns in response to a prompt.
Agentic AI	Takes a goal, breaks it down, and completes tasks with less human input using built-in tools.

The Core Mechanic

LLMs do not look things up. They predict the most probable next word based on everything given before it.

- **Vague prompt** = the model wanders through probability space and lands on a generic answer
- **Specific prompt** = the model takes a direct path and produces something actually useful
- **No memory between sessions** unless you provide context explicitly
- **Always verify the output.** It is not ground truth.

RAG: How Copilot Connects to Your Data

Retrieval Augmented Generation (RAG) is why Copilot can reference your actual emails, memos, and documents without being retrained. When you send a prompt, Copilot searches a pre-built index of your M365 environment, pulls the most relevant content, and adds it to your prompt before the model responds.

The practical implication: a vague prompt retrieves vague document chunks. A specific prompt retrieves better context. When Copilot gives you a bad answer, it is usually a retrieval problem, not a model problem.

2. The CRAFT Formula

CRAFT is a five-part structure that gives the model everything it needs to produce a useful output on the first try. Each element you include removes a decision the model would otherwise make on your behalf, and it will not always make the right one.

C Context: What does the AI need to know?

This is the background that gives body to the entire response. Skip it and the model invents its own assumptions, often wrong ones.

Weak	Strong
Summarize this earnings call.	I am reviewing this company for potential inclusion in our large-cap equity fund. This is their Q3 earnings call: they missed revenue consensus by 6% but beat on margins. Summarize the key themes relevant to a long-term equity holding decision.

R Role: Who should the AI act as?

Assigning a role shifts the model's expertise level, tone, register, and depth. It is the difference between a generic assistant and a specialist.

Weak	Strong
What do you think of this bond?	You are a fixed-income analyst evaluating a new corporate bond issuance for portfolio fit. Review this term sheet and flag anything that would concern a credit committee.

A Action: What exactly should the AI do?

This is the verb. The clearer and more specific it is, the shorter the model's inference path.

Weak	Strong
Look at these ETF documents.	Read these ETF prospectuses and pull out the expense ratio, underlying index, liquidity terms, and any tracking error disclosures for each one so I can compare them side by side.

F Format: How should the answer look?

Without format instructions, the model defaults to whatever is most statistically probable, which is usually prose when you wanted a table.

Weak	Strong
Summarize this 401k fund lineup.	Summarize this 401k fund lineup in a table with four columns: Fund Name, Asset Class, Expense Ratio, and Who It Is Suitable For.

T Target Audience: Who is this for?

This calibrates tone, depth, and vocabulary. The same information reads very differently for a credit committee versus a first-time investor.

Weak	Strong
Explain how futures hedging works in this portfolio.	Explain how the futures positions in this portfolio hedge against interest rate risk, as if you are briefing a client who understands they own bonds but has never seen derivatives before.

3. CRAFT in Action

Full Worked Prompt: Credit Committee Brief

I am reviewing a corporate bond issuance for potential inclusion in our fixed income fund. The issuer is a mid-cap industrial company and the coupon is 5.8% over 10 years. Act as a fixed income analyst preparing materials for a credit committee. Pull out the key credit risks, covenant terms, and how this bond compares to current treasury yields. Format as a structured one-pager with three sections: Credit Risk, Key Terms, and Committee Recommendation. This is for the credit committee; keep it concise and decision-ready.

	What each element is doing
C	Mid-cap industrial bond, 5.8% coupon, 10-year term, being reviewed for fixed income fund inclusion.
R	Fixed income analyst preparing for a credit committee.
A	Pull out credit risks, covenant terms, and comparison to treasury yields.
F	Structured one-pager with three named sections.
T	Credit committee. Concise and decision-ready.

Weak vs. Strong: College Savings Recommendation

Weak Prompt	Strong Prompt
Compare 529 plans for my client.	I have a client who is 35, has a 10-year-old child, lives in New York, and can contribute \$500 a month toward college savings. Act as a portfolio advisor specializing in education planning. Compare the New York 529 plan against two other highly-rated state plans, factoring in the state tax deduction my client would lose by going out of state, investment options, and fees. Format as a comparison table with a clear recommendation at the end. This is for an internal note before the client meeting, not client-facing.

The stronger prompt looks longer but requires fewer back-and-forth exchanges. Every CRAFT element removes a decision the model would otherwise make on its own.

Advanced Techniques, Organized by CRAFT

	Technique	What to do
C	Few-Shot Prompting	Include 2 to 3 examples of the input/output pattern you want before your actual request.
C	System 2 Attention	"First remove anything not relevant to my question, then answer." Strips noise from long inputs.
R	SimToM	Ask the model to identify each stakeholder's goals and constraints before responding. Useful for negotiations and multi-party communications.
A	Chain of Thought	Add "think step by step" before your request. The model shows its reasoning, which makes errors easier to catch.
A	Tree of Thoughts	Ask for multiple reasoning paths before picking the best one. Good for scenario planning and trade-off analysis.
F	Self-Evaluation	Ask Copilot to rate its own output and flag anything it is uncertain about before you accept it.
T	Emotion Prompting	Signal what is at stake. "This is going to a live client" raises the model's depth and care noticeably.

4. Prompting AI Agents

The same CRAFT framework applies to agents, but the consequences of a weak prompt are bigger. A bad chat answer is annoying. A bad agent prompt can trigger the wrong action on the wrong data in the wrong order and create a cascade of problems.

How CRAFT Shifts When You Are Prompting an Agent

	Element	In Chat	In Agents (extra layer)
C	Context	Background for the model	Also defines which data sources and systems the agent can access
R	Role	Sets tone and expertise	Also sets the agent's operating authority and limits what actions are appropriate
A	Action	Defines the task	Defines execution order; number each step explicitly
F	Format	Structures the output	Also specifies when the agent must pause and show output for review before continuing
T	Target	Calibrates tone and depth	Also specifies who the agent should escalate to when it is uncertain
S	Scope	Not needed in chat	Explicitly states what the agent may access, what it must skip, and when it must stop

The S stands for Scope. A missing scope element becomes a permission gap the agent fills with its own judgment. Add it every time.

Worked Example: Meeting Prep Agent

Step 1: Pull my calendar for tomorrow and list every external meeting. Step 2: For each external meeting, retrieve the last three emails I exchanged with that counterpart. Step 3: For each meeting, draft a three-bullet briefing covering context, last action item, and suggested talking point. Pause after Step 3 and show me all briefings before doing anything else. Scope: Calendar and Inbox only. Do not access SharePoint, Teams chats, or any shared drives.

Interactive: Build Your Own Morning Briefing

Use this scaffold to write a personalized analyst briefing in Copilot.

	Element	Fill this in
C	Context	Your role, team, and today's date
R	Role	What expertise should Copilot bring?
A	Action	What should it cover? (e.g. platform outages, settlement fails, regulatory news)
F	Format	How should it look? (e.g: Topic Key Development Who It Affects My Action Required)
T	Target	Your own morning prep. Flag time-sensitive items first. Add one personal section if you want.

Worked Example: Trading Systems and Settlement Operations

I work in trading systems and settlement operations at an asset management firm. Today is Monday. Act as a research assistant who understands both financial markets and market infrastructure technology. Give me a five-point morning briefing covering: any system outages or latency issues reported across major trading platforms, settlement fails or clearing house updates, and any regulatory or vendor news affecting trading technology. Format as a table: Topic | Key Development | Who It Affects | My Action Required. This is for my own morning prep; keep it brief and flag anything time-sensitive first. Also add one section at the end: World Cup, latest results and how Argentina performed in their most recent match.

Agent version: add this to C: "Access my firm's trading system logs, settlement reports, and relevant vendor communications only. Do not access any client data, personal files, or shared drives outside of operations. Do not send or forward anything. Generate a read-only output for my review."